

Kushagra Tiwary

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I'm a 3rd year PhD Student at MIT co-advised by [Boris Katz](#), [Tomaso Poggio](#), and [Ramesh Raskar](#). I also work closely with [Brian Cheung](#). My research aims to build artificial intelligence that can help explain scientific phenomena, and discover new and useful things inside worlds grounded by physics & simulations.

EDUCATION

Ph.D., Media Arts & Sciences, Massachusetts Institute of Technology 2023–Present

Advisor: [Boris Katz](#) · [Tomaso Poggio](#) · [Ramesh Raskar](#)

Interests: AI, AI for Science, Science of Intelligence, Neuroevolution, Simulations and Modeling

S.M., Media Arts & Sciences, Massachusetts Institute of Technology 2021–2023

Thesis: *Discovering, Learning, and Exploiting Visual Cues* ([Link](#)). Advisor: [Ramesh Raskar](#)

B.S., Electrical & Computer Engineering, University of Illinois Urbana-Champaign 2015–2019

B.S. With Honors. Focus on Computer Vision & Machine Learning.

PROFESSIONAL EXPERIENCE

Incoming Research Scientist Intern, Sakana AI Spring-Summer 2025

Will start collaboration with [Sebastian Risi](#) and [Brian Cheung](#).

Graduate Research Assistant, MIT Media Lab 2021–Present

Research Assistant working on understanding and building intelligence that can discover new things.

Visiting Student, UC Berkeley June 2025

Presented at [Redwood Seminar](#); visited [Bruno Olshausen's](#) group.

Graduate Student, BMM Summer Course (NSF) August 2025

One of thirty students selected for the program.

AI Engineer, Optimus Ride (Self-driving Startup) Aug 2018–Jun 2021

- Lead architect for Multi-Tasking Model and Traffic Light Deployment for nationwide fleet ([Patented](#))
- Lead release testing of models on next-gen vehicles in Boston
- Wrote training codebase; decreased training and release time by over 25%
- Designed company's first *Software 2.0* system to sample rare-events

SELECTED PUBLICATIONS

See [Google Scholar](#) for full list.

Journals

1. K. Tiwary*, A. Young*, ..., T. Poggio, D.-E. Nilsson, B. Cheung*, R. Raskar*. **What if Eye...? Computationally Recreating Vision Evolution.** *Science Advances*, 2025
[Paper](#) · [MIT News](#) · [Science News](#) · [Website](#) · [Hacker News](#) · [TEDx](#)
2. K. Tiwary, ..., T. Poggio, R. Raskar. **A Roadmap for Generative Design of Visual Intelligence.** *MIT Press: Impacts of Generative AI*, 2024

Selected Conference Papers

1. S. Longpre, ..., K. Tiwary (Video Lead), ..., S. Hooker, J. Kabbara. **Bridging the Data Provenance Gap Across Text, Speech and Video.** *ICLR 2025* [[arXiv](#), [MIT Tech Review](#), [Globe & Mail](#)]
2. Z. Tasneem, A. Dave, ..., K. Tiwary, ..., A. Veeraraghavan. **DecentNeRFs: Decentralized Neural Radiance Fields.** *ECCV 2024* [[Web](#), [arXiv](#)]
3. T. Klinghoffer*, K. Tiwary*, ..., R. Raskar. **DISeR: Designing Imaging Systems with RL.** *ICCV 2023* [[Web](#), [Paper](#)]
4. K. Tiwary*, A. Dave*, ..., R. Raskar. **ORCa: Objects as Radiance Field Cameras.** *CVPR 2023* [[Web](#), [arXiv](#), [MIT News](#)]

5. K. Tiwary*, T. Klinghoffer*, S. Somasundaram*, R. Raskar. **Physics vs. Learned Priors**. *ICCP* 2022 [[arXiv](#)]

Exhibitions

1. C. Chang, A. Young, K. Tiwary. **Recreating Vision Evolution**. [Museum of Science Boston](#). April 2025
2. C. Chang, A. Young, K. Tiwary. **First Signs of Vision**. [MIT Museum After Dark](#). April 2025

INVITED TALKS

What if Eye...? Computationally Recreating Vision Evolution

- Kempner Institute, Harvard December 2025
- [Redwood Seminar](#), UC Berkeley June 2025
- Caltech Vision Seminar June 2025
- [MIT Imagination in Action](#) April 2025
- Harvard Medical School March 2025
- CSAIL Vision Seminar February 2025
- [TEDx Boston](#) September 2024
- [New England Computer Vision Conference](#) (Oral) November 2024

Neural Rendering and Visual Cues (Multiple Papers)

- [MIT Imagination in Action](#) (Panel: Frontiers of AI Research) April 2024
- Hyundai Vision Conference, South Korea August 2023
- CSAIL Graphics Seminar September 2023

AWARDS, GRANTS, AND HONORS

- [Qualcomm Innovation Fellowship](#) (North America Winner) [[Media Lab](#), [EECS](#)] 2023
- [MIT-Israel Zuckerman STEM Fund Award](#) (\$30,000; 1 of 6 at MIT) 2022–23
- [MIT Generative AI Seed Grant](#) (1 of 16 at MIT) 2024
- Harold and Ruth Hayward Scholarship 2018

PATENTS

1. [Decentralized Neural Radiance Fields from Crowdsourced Images](#). US20250336144A1. 2025
2. [Efficient detection of structure and status of traffic lights](#). WO2022246412A1. 2022

SERVICE AND ORGANIZATION

Organizer: [MIT Visual Computing Seminar](#) (Fall 2025–Spring 2026)

Workshop Organizer: [Neural Fields Beyond Conventional Cameras](#) (ECCV 2024), [Accelerating Discovery](#) (MIT 2023)

Reviewing: TPAMI, CVPR, ECCV, ICCV, 3DV 2024, [GenAI for Design](#)

Mentoring: Nikhil Behari, Bhavya Agarwal, Katie Spivakovsky, Chaitanya Kapoor; [Media Lab SOS](#), [EECS GAAP](#)

SELECTED PRESS

- [Science News](#): “Why have so many different eyes evolved?” (Dec 2025)
- [MIT News](#): “Scientific sandbox lets researchers explore evolution of vision” (Dec 2025)
- [Globe & Mail](#), [MIT Tech Review](#): Data provenance coverage (Dec 2024)
- Consent in Crisis: [NYT](#), [404 Media](#), [Vox](#), [Yahoo](#) (2024)
- [TEDx Boston](#): “Can AI Recreate 500 Million Years of Vision Evolution?” (Sept 2024)
- [MIT News Front Page](#): “Using reflections to see the world” (May 2023)

SKILLS

Programming: Python, C/C++, TensorFlow 1.x/2+, PyTorch, OpenCV, TensorRT, ONNX, gRPC

Languages: English, Spanish (Professional Fluency—lived in Spain for 5 years), Hindi